


(a)

| Inputs | | Output (AND) | Output (NAND) |
|---|---|---|---|
| $A$ | $B$ | $Z = A.B$ | $Y = \overline{A.B}$ |
| 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

(b)

**Figure 10.44** (a)Two input NAND gate (b) Truth table

*Boolean equation:*

$$Y = \overline{A.B}$$

*Logic operation*

The output $Y$ equals the complement of AND operation. The circuit is an AND gate followed by a NOT gate. Therefore, it is summarized as NAND. The output is at logic zero only when all the inputs are high. The rest of the cases, the output is high (Logic 1 state). The truth table of NAND gate is shown in Figure 10.44(b).

## NOR gate

*Circuit symbol*

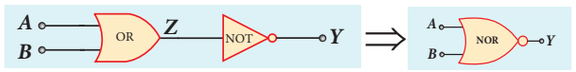
The circuit symbol of NOR gate is shown in Figure 10.45(a). $A$ and $B$ are inputs and $Y$ is the output.

*Boolean equation:*

$$Y = \overline{A + B}$$

*Logic operation*

The output $Y$ equals the complement of OR operation ($A$ OR $B$). The circuit is an OR gate followed by a NOT gate and is summarized as NOR. The output is high when all the inputs are low. The output is low for all other combinations of inputs. The truth table of NOR gate is shown in Figure 10.45(b).




(a)

| Inputs | | Output (OR) | Output (NOR) |
|---|---|---|---|
| $A$ | $B$ | $Z = A+B$ | $Y = \overline{A+B}$ |
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 |

(b)

**Figure 10.45** (a) NOR gate (b) Truth table

## Ex-OR gate

*Circuit symbol*

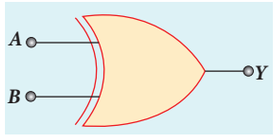
The circuit symbol of Ex-OR gate is shown in Figure 10.46(a). $A$ and $B$ are inputs and $Y$ is the output. The Ex-OR operation is denoted as $\oplus$.

*Boolean equation*

$$Y = A.\overline{B} + \overline{A}.B$$
$$Y = A \oplus B$$

*Logic operation*

The output is high only when either of the two inputs is high. In the case of an Ex-OR gate with more than two inputs, the output will be high when odd number of inputs are high. The truth table of Ex-OR gate is shown in Figure 10.46(b).




(a)

| Inputs | | Output (Ex-OR) |
|---|---|---|
| $A$ | $B$ | $Y = A \oplus B$ |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(b)

**Figure 10.46** (a) Ex-OR gate (b) Truth table



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